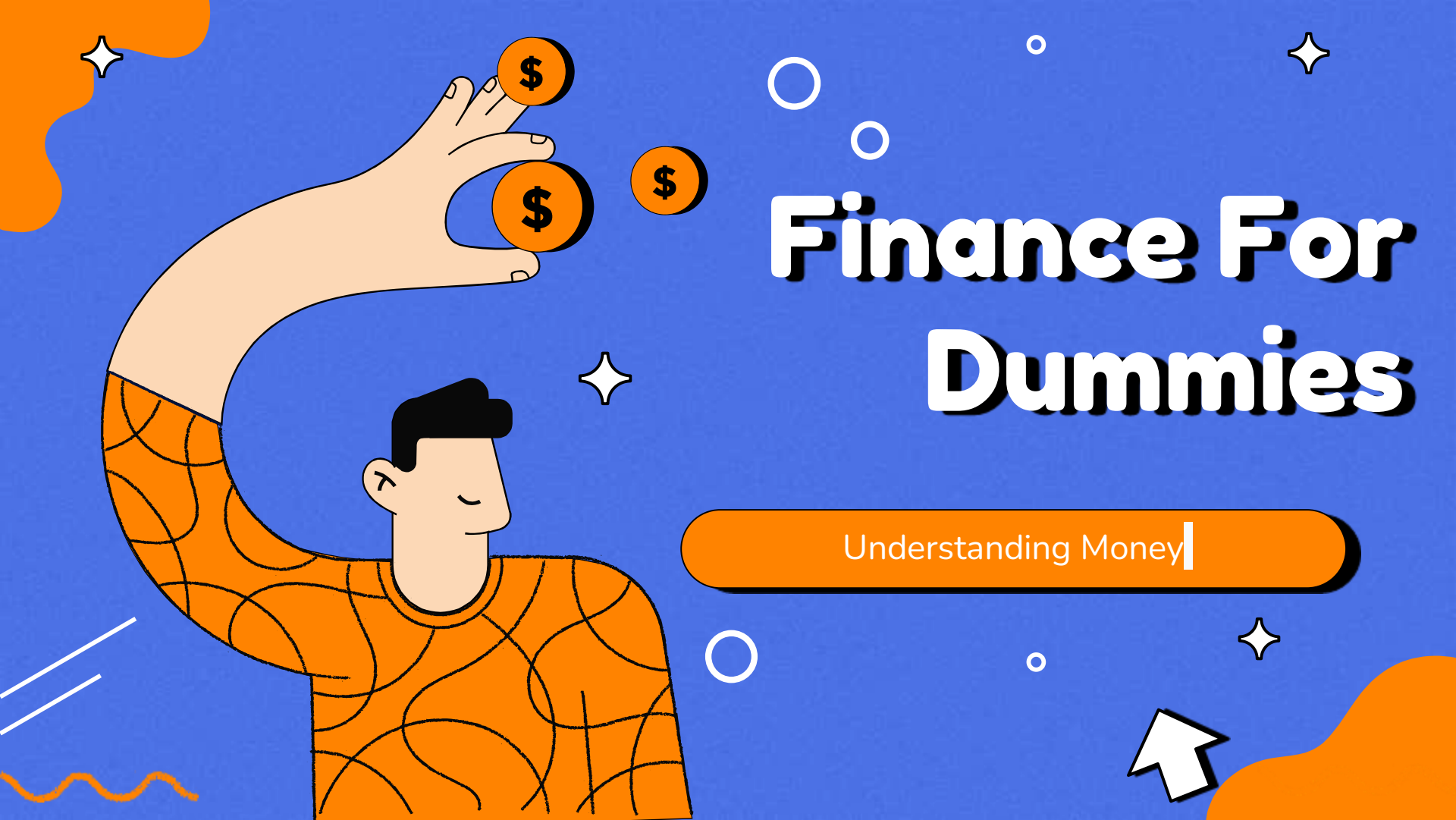
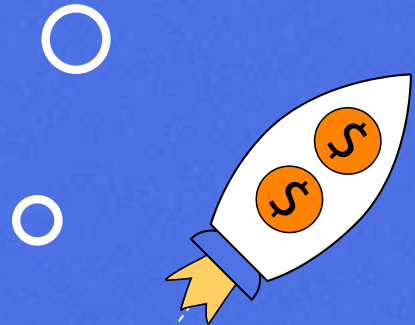




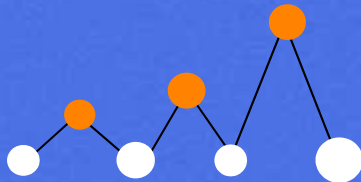
Finance For Dummies

Understanding Money |



“Money is a Human invention, Like
Language, and is subject to human
ingenuity and Human error”

—**Milton Fried**





Before Money was invented

Barter System is a method of trade in which goods or services are exchanged directly for other goods or services





Problems with Barter System



01

Double Coincidence of wants

02



Lack of common measure of Value

03

Difficulty in storing wealth

04



Limited Scope for specialisation

05



Lack of standardisation





Money As Technology

Money solves the problem of the barter system by providing a common denominator for the exchange of goods and services, allowing for more efficient trade and the ability to store value.



Different Forms of Money



**LiveStock in
Mesopotamia**



Grains in EGYPT

Different Forms of Money



Shells



Gold Coins



Gold Coins became the most useful forms of Money for many centuries



Different Forms of Money



Paper Money



Cryptocurrency???



Properties of Money



01

Medium of exchange

02



Unit of account

03

Portable

04

Durable

05

Divisible

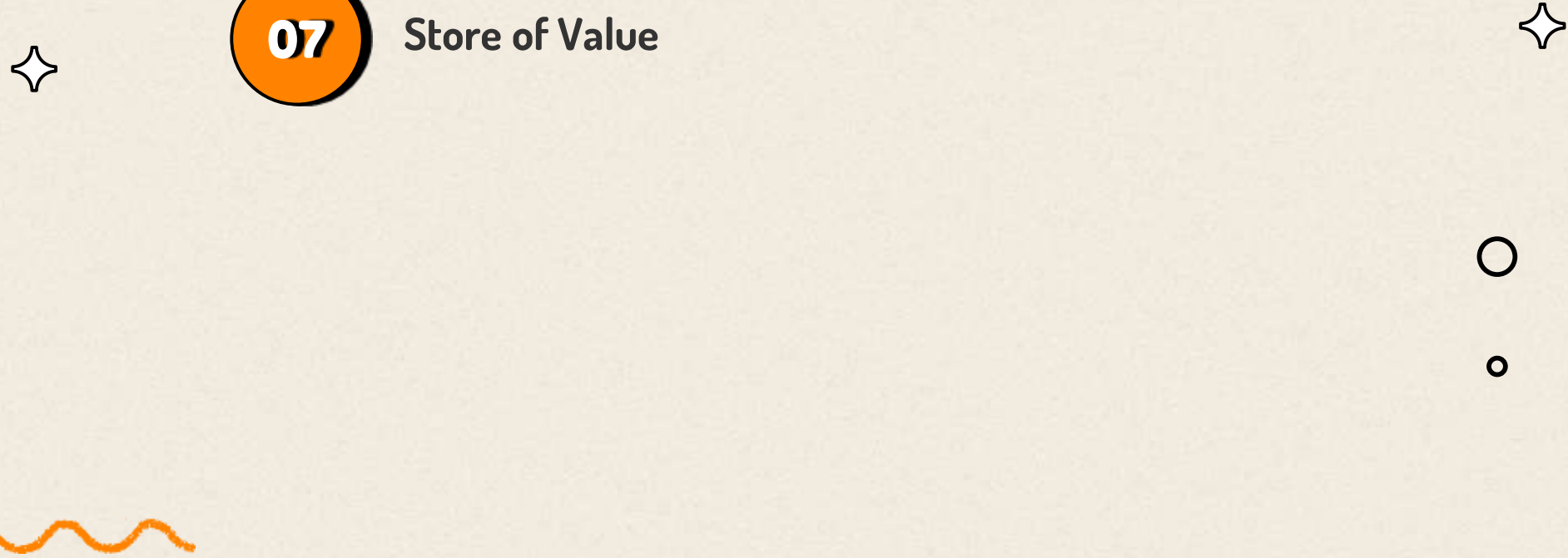
06

Fungible





Properties of Money



07

Store of Value



Gold as Money

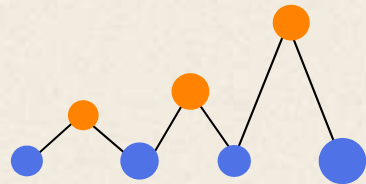
Started 5000 years ago



Gold as Money

5000 years Ago Egyptians started using Gold and silver

- Almost Entire world was using Gold coins as currency in the first 1000 years of AD.
- Though coins each country used was different from one another, yet GOLD standard was widely popular for exchanging value.





Roman Empire

The Roman Empire was one of the greatest and most influential civilisations in the world and lasted for over a 1000 years.



Roman Empire

Roman Empire was on Gold standard but after a lot of Military overexpansion and military spending they started debasing their Gold coins. First they add small amount (5-10%) of bronze in Gold coins which later increased to 80-90%.

- Eventually Traders stop accepting that currency and the entire economy collapsed.
- Gold, silver, and various forms of local and regional currencies
- Some Communities resorted to bartering



Money is Currency+Store of Value

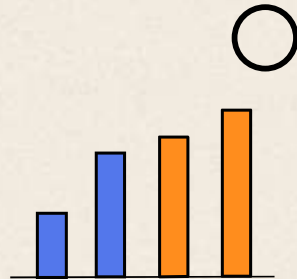
Store of Value is must for currency



Why Gold Survived as Money?



1. IT IS ULTIMATE MONEY BECAUSE THERE IS NOTHING IN THE SAME LEAGUE IT IS DIVISIBLE
2. PERMANENT STORE VALUE
3. IT DOESN'T GO AWAY
4. IT KEEPS GOVERNMENT UNDER CONTROL. A SOLVENT SYSTEM
5. THEY CAN'T SPEND BEYOND ITS MEANS IN GOLD AS CURRENCY
6. THEY CAN'T PRINT AS MUCH AS THEY WANT





02

Paper Money

Started in 618-907 AD



Invention of Printing Press and Shift in currency

Printing technology, in the form of movable type, was invented by the Chinese during the Tang Dynasty (618-907 AD).



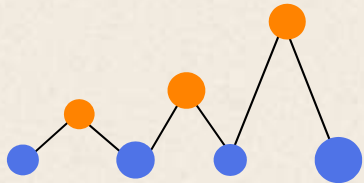
Paper Currency

The first known use of paper money was in China, during the Tang Dynasty (618-907 AD). The Chinese used paper money in the form of "jiaozi," which were promissory notes issued by the government that could be exchanged for coins.

- Paper money was also used in medieval Europe, in the form of "bills of exchange," which were promissory notes that could be used to facilitate trade.
- These bills of exchange could be used to transfer funds between merchants, or to pay for goods and services, without the need to physically transport coins.

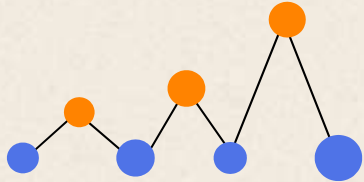
Transition from Gold to Paper

- Convenience: Paper money is easier to carry and transport than gold coins, which made it a more convenient form of currency.
- Government issuance: Governments began issuing paper money as a way to finance wars and other expenditures. The government would issue promissory notes, which were promises to pay the bearer a certain amount of gold upon demand. These promissory notes, or paper money, were widely accepted as a form of payment, and they eventually replaced gold coins as the dominant form of currency.
- Limited supply of gold: There is a limited supply of gold, which makes it difficult to produce enough gold coins to meet the demand for currency. Paper money, on the other hand, can be printed in larger quantities as needed.



Transition from Gold to Paper

- Countries were using Gold coins as currencies Initially
- To increase convenience of trades and increase spending they started issuing Paper currencies that was claimed to be backed by gold. This type of money is called Commodity Money
- By 1930s during the great depression, Almost all the countries abandoned Gold standard and started using Paper currencies





Fiat Currencies

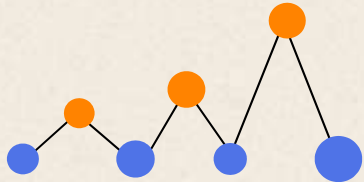
Present Form of Money

What is Fiat Money?

- Fiat money is a type of currency that is issued by a government and is not backed by a physical commodity, such as gold or silver.
- The value of fiat money is derived from the relationship between supply and demand and the stability of the issuing government, rather than the value of a commodity backing it.
- Fiat money is different from commodity money, which is a type of money that is backed by a physical commodity, such as gold or silver. Commodity money is valued based on the value of the commodity that backs it, whereas fiat money is valued based on the issuing government's ability to maintain its value.

How do Fiat is stabilised?

- Monetary policy: Governments can use monetary policy, which involves the management of the supply and demand of money, to stabilize the value of their currencies. For example, a central bank may raise or lower interest rates in order to affect the supply of money and credit in the economy, which can help to stabilize prices and the value of the currency.
- Fiscal policy: Governments can also use fiscal policy, which involves the management of government spending and taxation, to stabilize the value of their currencies. For example, a government may increase or decrease spending or change tax rates in order to affect demand in the economy and help to stabilize prices and the value of the currency.



How do Fiat is stabilised?

- Exchange rate policy: Governments can use exchange rate policy to stabilize the value of their currencies. For example, a government may peg its currency to the value of another currency, such as the US dollar, in order to stabilize its value.
- Economic reforms: Governments can implement economic reforms, such as liberalizing markets or improving the business climate, in order to increase economic growth and stability, which can help to stabilize the value of the currency.

**Real wealth is Time
and Freedom**



Technology Adoption Curves

How do we know if a technology is real?



What is Technology Adoption Curve?



The technology adoption curve is a model that describes the stages of acceptance and usage of a new technology.





Why is it Important?



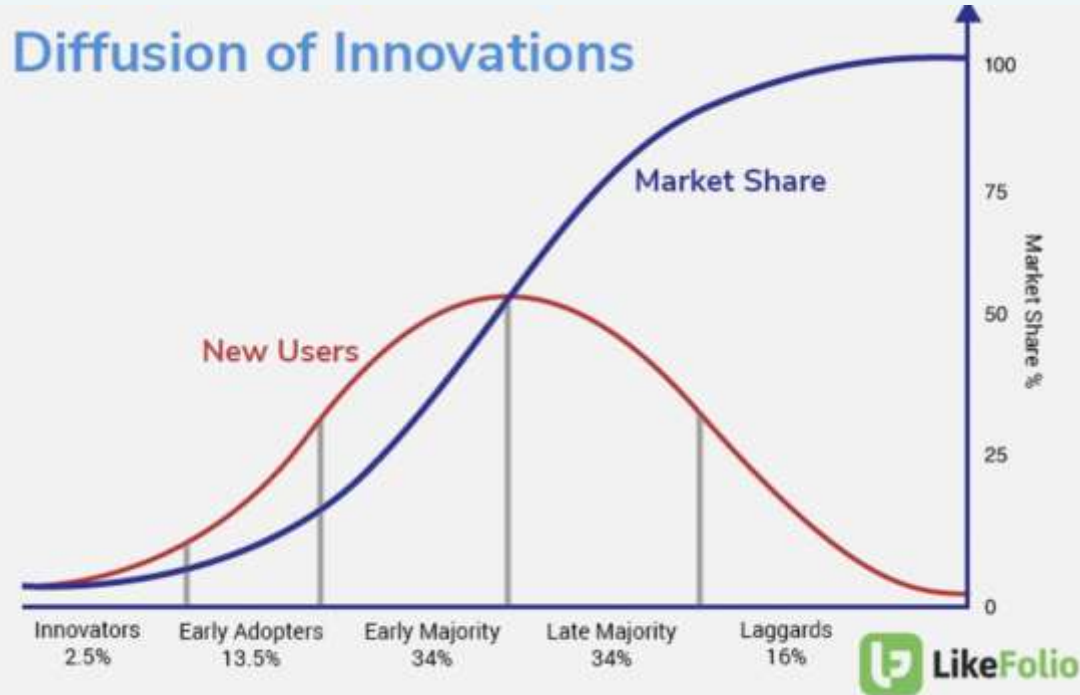
It is important to understand the technology adoption curve as it provides a framework for understanding how new technologies are adopted and how they can be used to address societal and business needs. Additionally, it provides insights into the factors that influence technology adoption, including cultural and social factors, technological capabilities, and the availability of resources.



Diffusion of Innovations

Diffusion of innovations is a theory explaining how, why and at what rate new ideas and technology spread along the adoption curve

Diffusion of Innovations



Innovators

The first users to try something are represented by a small slice of the population called Innovators, ready to try the next new thing at the drop of the hat.

As time moves on, if the technology is successful, more and more people adopt it.

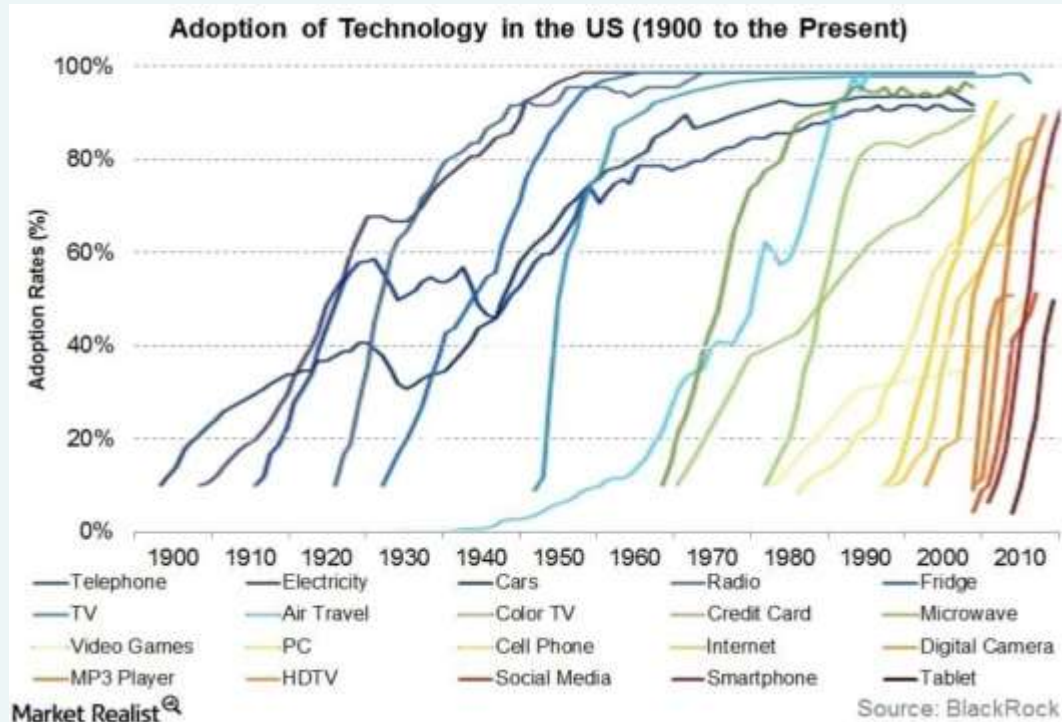
Early Adopters

Early Adopters are a small but critical group to the success of a technology's spread. If a technology can succeed with that section of the population, it will hit a tipping point.

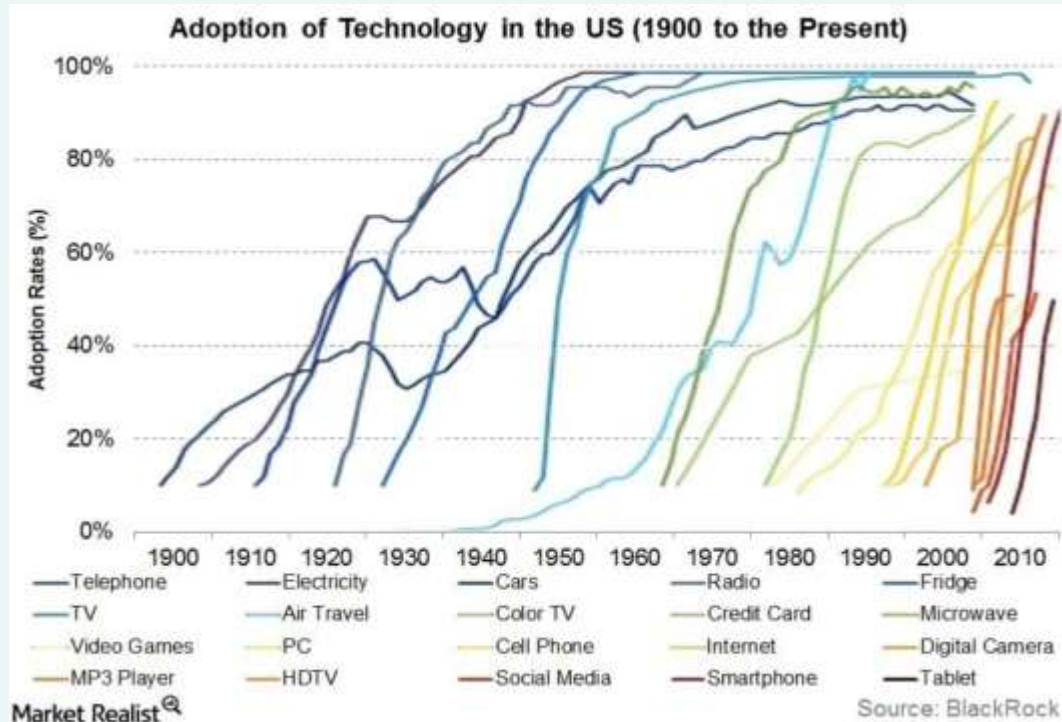
Early Adopters

That's where things get really interesting, as the next two stages involve the majority of the population adopting the technology, crossing 50% market share (purple line above).

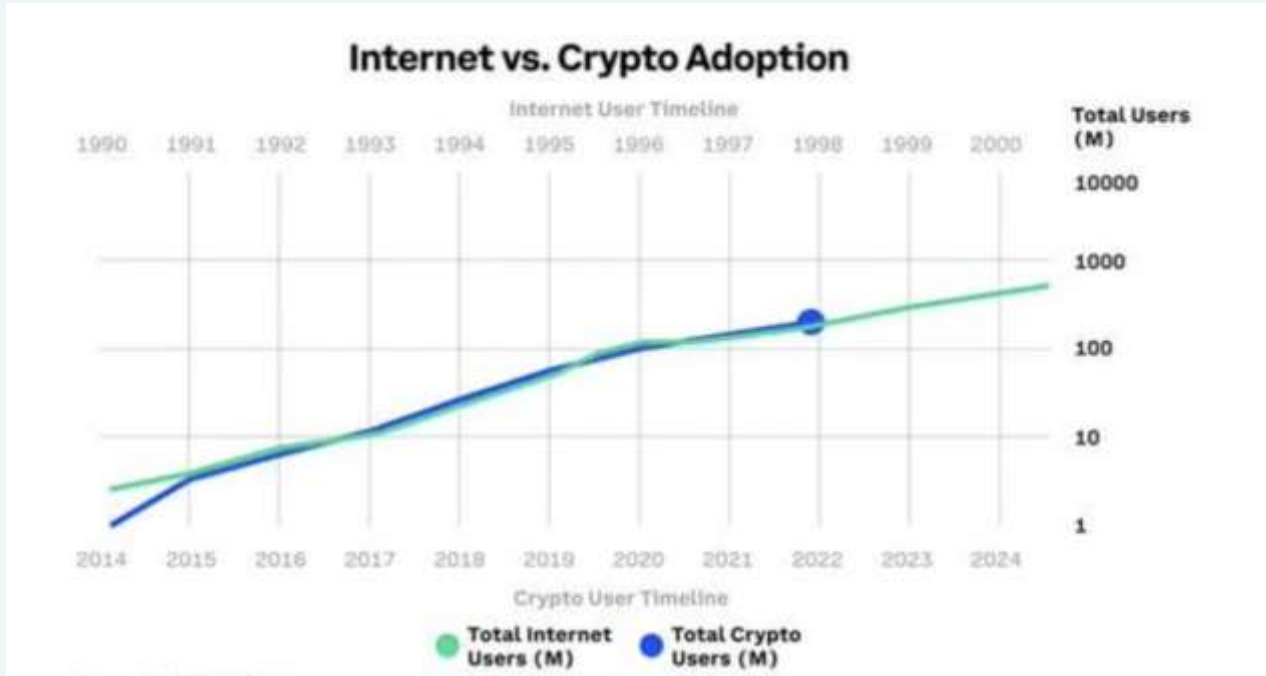
Different Technologies



Different Technologies



Internet Vs Crypto



6 Ds for Tech Disruption

Digitisation

Once technology digitizes into code and data, we can use computers to manage its applications.

Deception

Exponential growth starts out almost imperceptible at first before rapidly progressing.

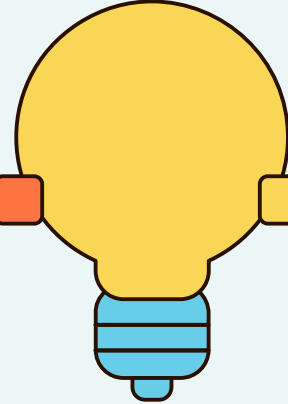
Disruption

Exponential technology can shatter previous paradigms and systems, rendering them obsolete.

01

02

03



04

05

06

Dematerialization

Due to the advances in microsensors, tangible technology can transform into digital apps.

Demonetization

When technology becomes digitized, the cost of duplication becomes near zero

Democratization

Products and services previously used by a privileged few become accessible to a large part of the global population, enabling massive change.

6 Ds for Tech Disruption

Leaders who see where technology is developing on this roadmap can capitalize on unrealized potential and innovate over their competition.

Case Study

Zoom Video Communication

Here's what that looked like for Zoom from 2014 through the end of 2019:

The company was clearly growing its user base and moving nicely along the adoption curve.



Chart shows number of people talking about zoom (Social Media mentions)

Case Study

Zoom Video Communication

At the end of 2019, Zoom was probably about here on the adoption curve:



Case Study

Zoom Video Communication

And then the COVID-19 crisis hit.

All of a sudden, entire populations are in quarantine and a majority of businesses, schools and organizations are trying to figure out how to allow their people to connect with each other remotely.

Because of their success with Innovators and Early Adopters, they garnered the Opinion Leadership necessary to be THE solution that people recommended when organizations were looking for video conferencing solutions.

Case Study

Zoom Video Communication

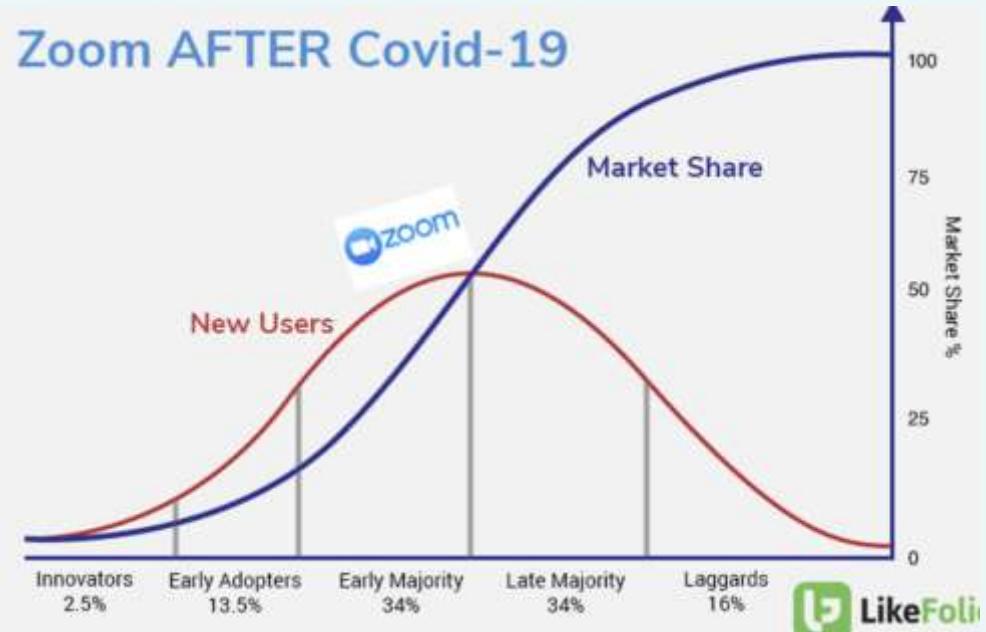
The “really nice growth curve” for Zoom prior to the crisis almost disappears because the adoption rate grew by 1,700% in a matter of weeks.



Case Study

Zoom Video Communication

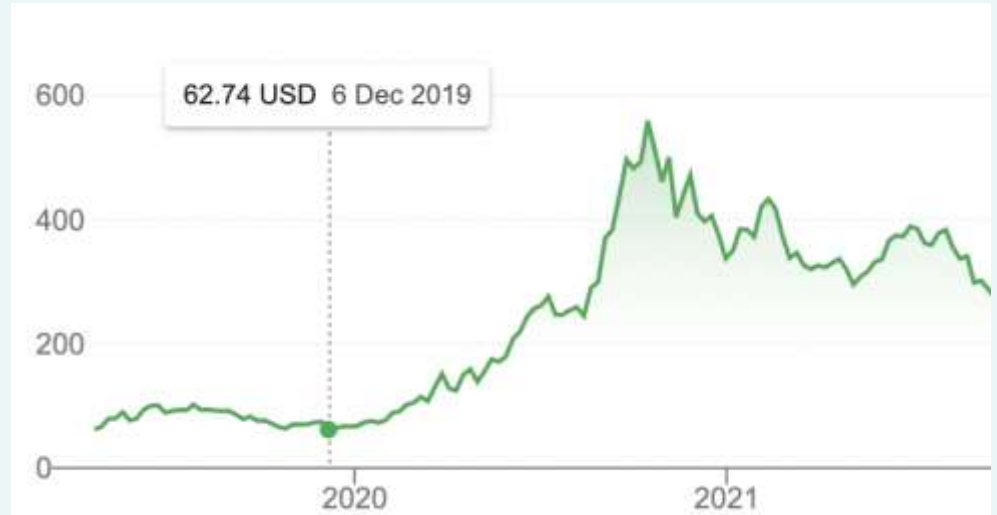
ZOOM on the adoption curve now:



Case Study

Zoom Video Communication

Zoom Share
performance in
Market



Case Study

Zoom Video Communication

The COVID-19 crisis has allowed Zoom to move all the way from the beginning of the Early Adopters stage to the middle of the Majority of the growth curve.

And it's made this giant move in months, while the move from Innovators to Early Adopters had taken years.

Lesson?

Identifying the companies that are doing so very quickly and at an accelerating pace can lead to tremendous, life-changing profits for investors in a short amount of time.

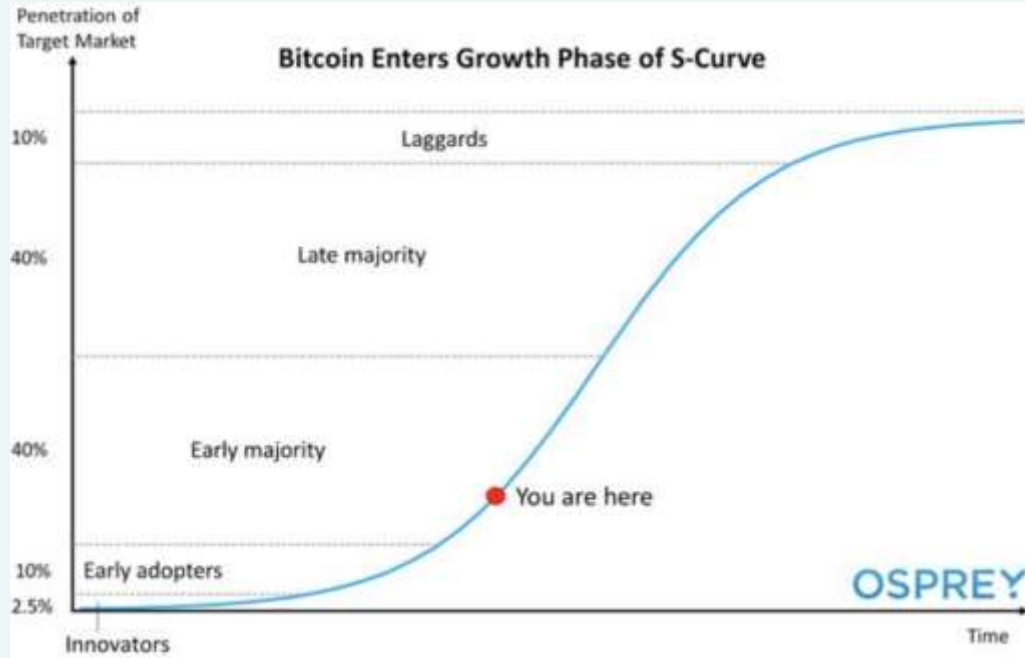
Blockchain on Technology adoption Curve?

How Common Is Crypto?

Share of respondents in selected countries who said that they used or owned crypto currencies (2020)



Blockchain on Technology adoption Curve?



Advantages for Companies riding Technology Adoption Curve?

Reduced competition: Companies that get to majority market share quickly benefit from a large technology and branding moat behind them.

More revenues: The more people use your product, the more money comes in the door. Capturing “Early Adopters” means 6.4x more users (and revenue) than only having the “Innovators”, and getting the Early Majority triples your user base from there.

Less marketing expenses: Companies that accelerate through the adoption curve usually do so because of successful network effects built into their product and good old-fashioned word of mouth. Opinion Leaders are doing their marketing for them, leaving more cash in investors’ hands.

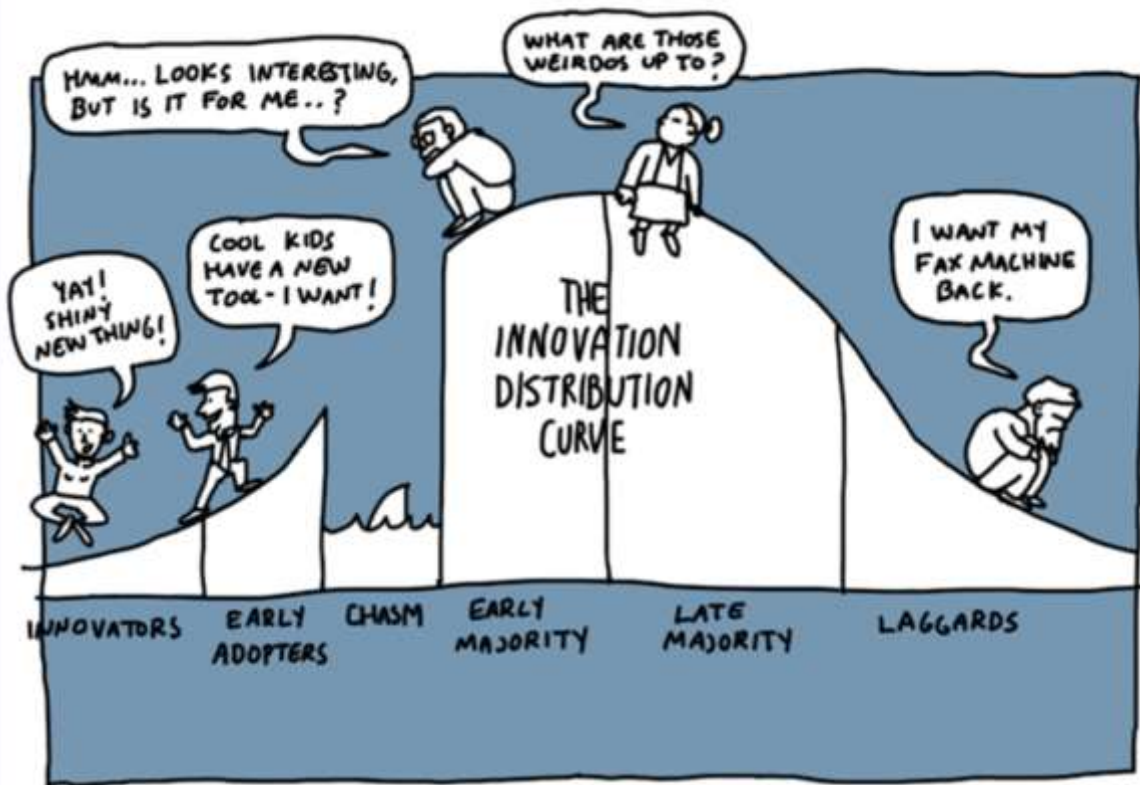
Growth becomes automatic: There are huge risks of failure at the Innovator and Early Adopter phases of adoption, but once a company has made it through the Early Majority phase, new user growth almost becomes a self-fulfilling prophecy. Think about how easy it was for Facebook to add users 250 million -300 million compared to users 1 – 50 million. “Everyone is on it, so I’ll join too.”

How to Spot Companies Accelerating Through The Adoption Curve

One way to figure out if a company is accelerating through the adoption curve is to simply keep an eye on it.

Innovators and early adopters are pretty good at spotting new technologies and watching their adoption rates among friends, family and businesses they associate with

Participants Reactions



Product Adoption

How to Improve Product Adoption



Improve your product



Make your support more supportive



Invest in user onboarding

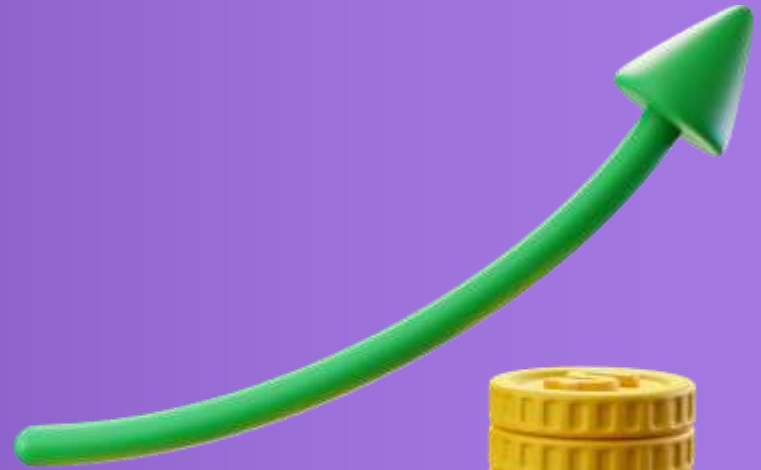


Engage with users in-product



Engage with users off-product

Assets and Financial Products



Commonly Traded Assets

```
graph TD; A[Commonly Traded Assets] --> B[Securities]; A --> C[Currencies]; A --> D[Contracts]; A --> E[Commodities]; A --> F[Real Assets]; B --> G[Fixed Income]; B --> H[Equities]; B --> I[Pooled Investments]; D --> J[Forwards]; D --> K[Futures]; D --> L[Swaps]; D --> M[Options]; D --> N[Others];
```

Securities

Currencies

Contracts

Commodities

Real Assets

Fixed Income

Equities

Pooled Investments

Forwards

Futures

Swaps

Options

Others



01

+ Securities

Fixed Income Securities

Promise to pay borrowed money

Include predetermined payment schedules, which usually include interest and principal payments

Can be short-term, intermediate-term, and long term

Examples- Bonds, notes, bills, certificates of deposit, commercial paper, repurchase agreements, loan agreements, mortgages

Equities

Ownership rights in companies

Include common shares, preferred shares, warrants



Pooled Investments

People invest in them to benefit from diversification and investment management services

Can be open-ended or closed-ended

Examples- mutual funds, trusts, depositories, hedge funds

Currencies

Some currencies are regarded as reserve currencies- are held by national central banks and other monetary authorities

Primary reserve currencies are US dollar and euro

Trade in foreign exchange markets

Contracts

Contracts are agreements among traders in the future



Value depends on the value of the underlying asset, which may be commodity, security, index, currency pair

Can be physical settled or cash settled

Forwards- Agreement to trade the underlying asset in the future at a price agreed upon today. Counterparty risk and liquidity problem limit the usefulness of forward contracts



Contracts



Futures- Standardized forward contract in which clearinghouse takes the guarantee

Swaps- Agreement to exchange payments of periodic cash flows that depend on future asset prices or interest rates

Others- Insurance contracts pay their beneficiaries a cash benefit if some event occurs, example being credit default swaps (CDS)



Option Contract

Option trading is a financial instrument that allows traders to speculate on the future price movements of an underlying asset, such as a stock, commodity, currency, or index.



Options are contracts that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price (strike price) within a specified time frame.

Options traders can use a variety of strategies, including buying calls (the right to buy the underlying asset at a specific price) or puts (the right to sell the underlying asset at a specific price), or using combinations of calls and puts to create more complex strategies.

Option trading is considered a more advanced form of investing and requires a good understanding of the markets and the underlying assets.



Terms in Option trading

- 
1. **Option:** A contract that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price (strike price) within a specified time frame.
 2. **Underlying asset:** The asset that the option contract is based on, such as a stock, commodity, currency, or index.
 3. **Strike price:** The price at which the option gives the buyer the right to buy or sell the underlying asset.
 4. **Expiration date:** The date on which the option contract expires and can no longer be executed.
 5. **Call option:** An option that gives the buyer the right to buy the underlying asset at the strike price
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- 

Terms in Option trading

6. Put option: An option that gives the buyer the right to sell the underlying asset at the strike price.



7. Premium: The price that the option buyer pays for the option contract.

8. In-the-money: An option that has intrinsic value, meaning that it has the potential to be profitable if exercised.

9. Out-of-the-money: An option that has no intrinsic value and will only be profitable if the underlying asset price moves significantly in the desired direction.

10. Exercise: The act of executing the right granted by the option contract, either to buy



Commodities

- Include precious metals, energy products, agricultural products
- Can be spot markets or forward and futures markets
- Producers and processors of industrial metals and agricultural products are the primary users of spot commodity markets



Real Assets

- Include tangible properties such as real estate, machinery
- Market is very illiquid because no two assets are alike, and managing them is costly, hence complicating the valuation of real assets
- Investors seeking exposure to real assets can buy shares of companies that hold and operate them





Questions For Understanding



Q1

A trader buys a stock for Rs.100. The leverage ratio is 2.5. A year later the trader sells the stock for Rs.120. What is the trader's return for the year?



Q1 A trader buys a stock for Rs.100. The leverage ratio is 2.5. A year later the trader sells the stock for Rs.120. What is the trader's return for the year?

+Ans

Return on unleveraged position: $(120-100)/100 = 20\%$

Therefore, return on leveraged position: $20 \times 2.5 = 50\%$



Q2

The current price of stock XYZ is Rs.250. An investor has bought a call option on the share of XYZ for a premium of Rs.7. The call option expires in 6 months and has an exercise price of 270. On the expiration date, the investor will exercise the call option if the share price of XYZ is:

- a) Below Rs.270
 - b) Above Rs.270
 - c) Above Rs.277
 - d) Below Rs.277
- 



Q2

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- a) Below Rs.270
- b) Above Rs.270
- c) Above Rs.277
- d) Below Rs.277

Ans

Above 270





Q3

Stock ABC's futures contract specifies delivery of 10 shares when contract settles. Clearing house requires its clients to post an initial margin of Rs.500 per contract and maintenance margin of Rs.400 per contract. A client buys 100 contracts at Rs.7500 per contract. The next day, the contract settles for Rs.7300 per contract. How much additional margin will the client have to provide to the clearinghouse?





Q3

Stock ABC's futures contract specifies delivery of 10 shares when contract settles. Clearing house requires its clients to post an initial margin of Rs.500 per contract and maintenance margin of Rs.400 per contract. A client buys 100 contracts at Rs.7500 per contract. The next day, the contract settles for Rs.7300 per contract. How much additional margin will the client have to provide to the clearinghouse?

Ans

Rs.200 Amount lost per contract: $7500 - 7300 = 200$ Total lost: $100 * 200 = 20,000$
Initial margin of 50,000 ($500 * 100$) is reduced to 30,000 ($50,000 - 20,000$), which is below the maintenance margin of 40,000 ($400 * 100$) Account has to be replenished to the initial margin





Q4

In an underwritten offering, the risk that the entire issue may not be sold to the public at the offering price is borne by:

- a) Issuer
 - b) Investment bank
 - c) Buyers that have bought the issue
- 



Q4

In an underwritten offering, the risk that the entire issue may not be sold to the public at the offering price is borne by:

- a) Issuer
- b) Investment bank
- c) Buyers that have bought the issue

Ans

b) Investment bank





Q5

Which of the following is least likely to be an objective of market regulation?

- a) Minimize agency problems
- b) Control fraud
- c) Ensure investors achieve a return that is equal to or higher than the risk free rate of return
- d) Promote fairness

Ans

c) Ensure investors achieve a return that is equal to or higher than the risk free rate of return



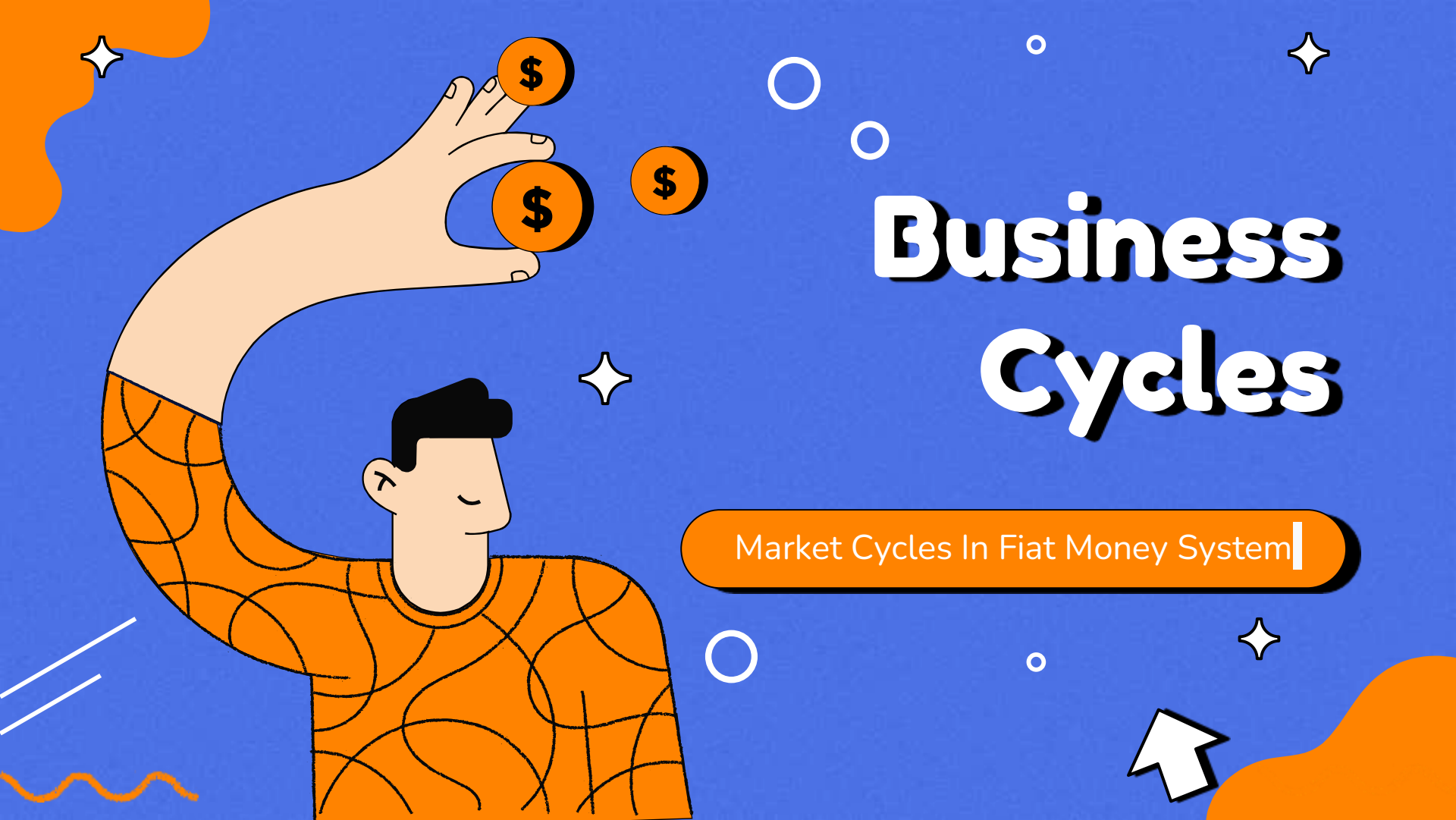


Q5

Which of the following is least likely to be an objective of market regulation?

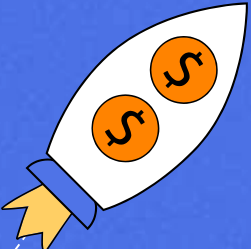
- a) Minimize agency problems
- b) Control fraud
- c) Ensure investors achieve a return that is equal to or higher than the risk free rate of return
- d) Promote fairness





Business Cycles

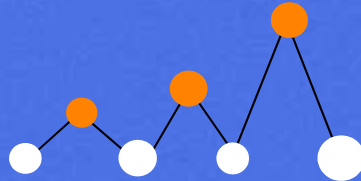
Market Cycles In Fiat Money System |



“The market is a pendulum that forever swings between unsustainable optimism (which makes stocks too expensive) and unjustified pessimism (which makes them too cheap)”

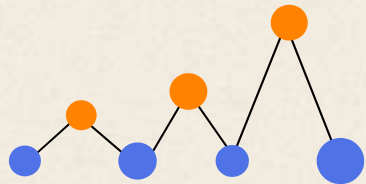


-Howard Marks



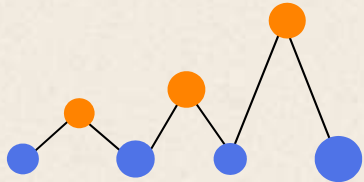
What is Market Cycles?

Market cycles refer to the recurring patterns of growth and decline that occur in various financial markets. These cycles can be observed in the stock market, the housing market, and the broader economy.



What is Market Cycles?

The most well-known market cycle is the business cycle, also known as the economic cycle. The business cycle is characterized by four phases: expansion, peak, contraction, and trough.





01

Expansion Phase

Period of Economic Growth/ Bull Markets





Characteristic of Expansion Phase



01

Increase in economic activity

02

GDP and Employment are rising



03

Increase in Consumer spending and business investment

04

Stock market tends to perform well



05

Inflation is generally low

06

Interest rates are often low





Characteristic of Expansion Phase



07

Unemployment is typically low



08



Consumer confidence is high



09

Businesses are optimistic



10



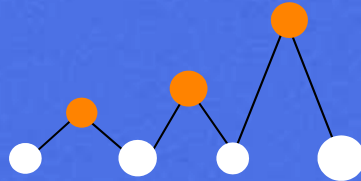
Can last for several years



“Expansion means taking risks, but the bigger the risk, the bigger the reward”



–Robert Kiyosaki





02

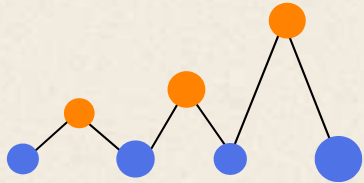
Peak of market Cycle

Euphoria Phase



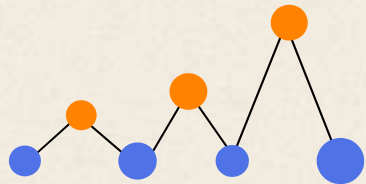
What is Market Cycle Peak?

At the peak of the market cycle, the economy has reached its maximum level of growth. This means that the level of economic activity, employment, and income are at their highest point, and the economy is operating at or near full capacity



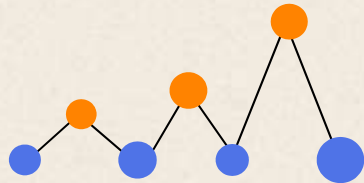
What is Market Cycle Peak?

In the stock market, the peak is the point at which stock prices are at their highest level and investors are the most optimistic. The market is considered to be overbought, as prices are high relative to the underlying fundamentals. Investors may begin to take profits, and the market may start to correct itself.



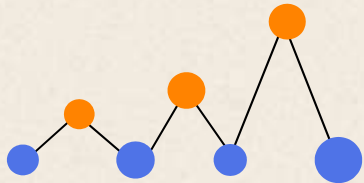
What is Market Cycle Peak?

At this stage, inflation may start to rise as the economy is operating at full capacity and there is more demand for goods and services, which can push prices up. This may cause central banks to increase interest rates to curb inflation.



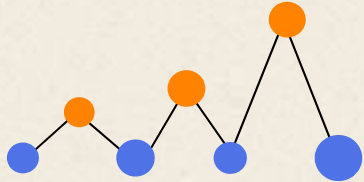
What is Market Cycle Peak?

Unemployment is typically low at the peak of the market cycle, as businesses are hiring to meet the increased demand for goods and services.



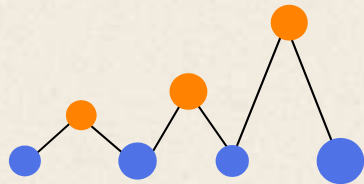
What is Market Cycle Peak?

Consumers are confident and spending, and businesses are optimistic and investing. This creates a virtuous cycle of growth, but it also increases the risk of a recession in the future as the economy may become overheated.



What is Market Cycle Peak?

The peak of a market cycle is a tricky point to identify, as the economy and stock market can continue to grow for some time before the peak is reached.





03

Contraction Phase

Declining Economic Activity/Bear Market





Characteristic of Expansion Phase



01

A contraction phase, also known as a bear market, is a period of declining economic activity characterized by falling asset prices, particularly in the stock market.



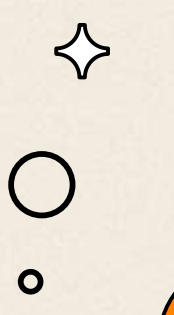
02

During a bear market, investors tend to become more risk-averse and pull their money out of the stock market, causing prices to fall further.

03

Economic indicators such as gross domestic product (GDP) and employment tend to decline during a bear market.





Characteristic of Contraction Phase

04

Consumer spending, business investment, and international trade also tend to decline during a bear market.



05

A bear market can be a difficult time for investors, as it can take several years for stock prices to recover to their previous levels.

06

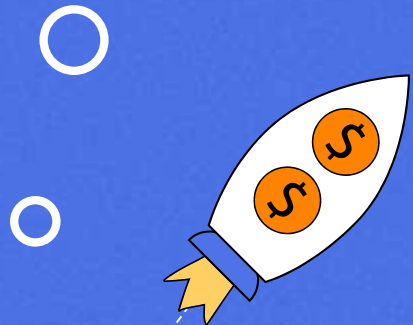
A bear market is typically defined as a decline of 20% or more in stock market indexes, such as the S&P 500.



07

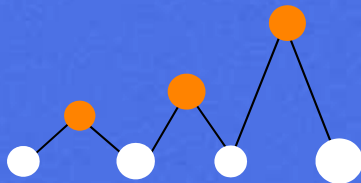
A decline in GDP for two consecutive quarters is often considered as a sign of a recession.





“A bear market is the time to be patient, not the time to panic”

—Peter Lynch





04

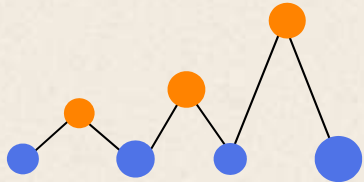
Trough

Point of maximum pessimism



What is Trough?

The trough is the point at which the economy reaches its minimum level of activity. At this point, the stock market is typically at its lowest point, and the economy is operating well below full capacity.





Characteristic of Trough



01

Low GDP growth: The trough of an economic cycle is characterized by a period of low GDP growth, as economic activity slows down.

02

Rising unemployment: As economic activity slows, companies may lay off workers, leading to higher unemployment rates.

03

Reduced consumer spending: As people lose their jobs or see their income decline, they may cut back on their spending, further weakening the economy.





Characteristic of Trough



04

Low business investment: As economic conditions weaken, businesses may be less willing to invest in new projects or expand their operations.

05

Reduced credit availability: Banks and other lenders may be less willing to extend credit to businesses and consumers, further slowing economic activity.

06

Low inflation or deflation: As economic activity slows, prices for goods and services may also decline, leading to low inflation or even deflation



07

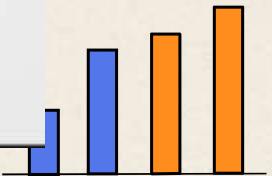
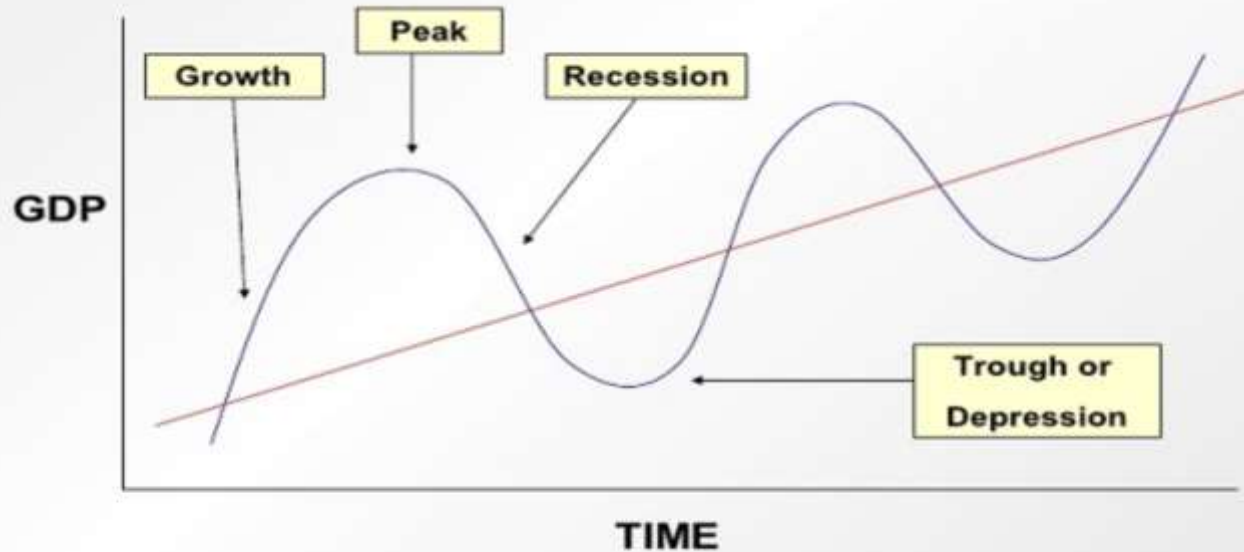
Central bank intervention: Central banks may take steps to stimulate the economy and promote recovery, such as cutting interest rates or launching quantitative easing programs.





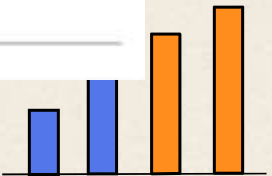
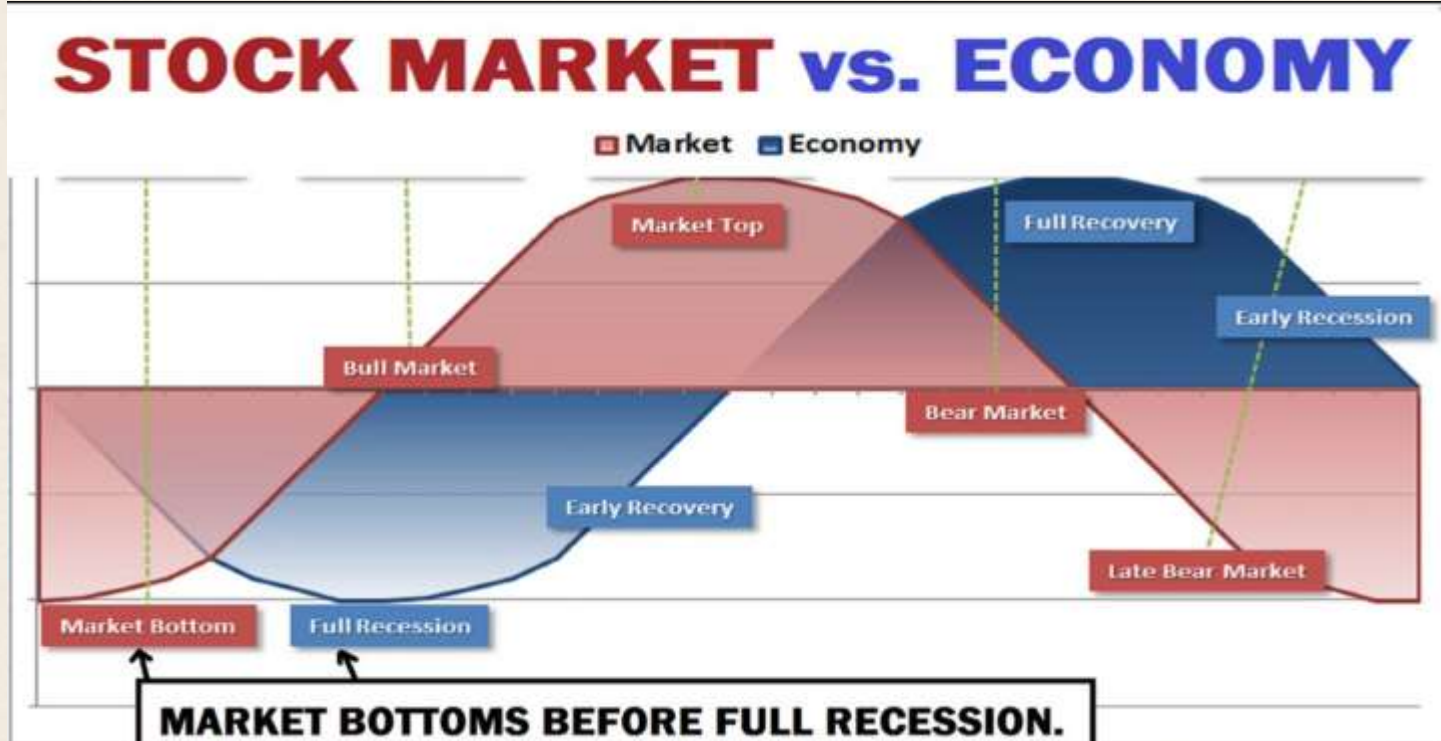
Economic Cycles

The Business Cycle





Economy vs Stock Market





Impact of Market Cycles

Market cycles can have a significant impact on individuals and society as a whole. Some of the ways in which market cycles may impact us include:





Impact of Market Cycle



01

Investment returns: During a market upturn, investments such as stocks may perform well and generate higher returns for investors. However, during a market downturn, investments may lose value, leading to losses for investors



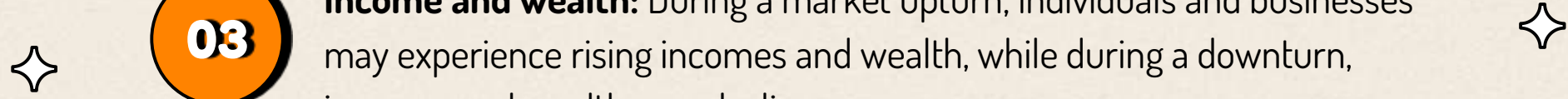
02

Employment: During a strong economy, employment opportunities may be plentiful, while during a recession, unemployment rates may rise and job opportunities may be scarce.





Impact of Market Cycle



03

Income and wealth: During a market upturn, individuals and businesses may experience rising incomes and wealth, while during a downturn, incomes and wealth may decline.

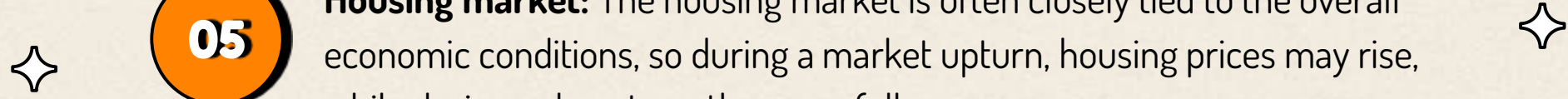
04



Consumer spending: During a strong economy, consumer spending may be strong, driving economic growth. However, during a recession, consumer spending may decline, further slowing economic activity.

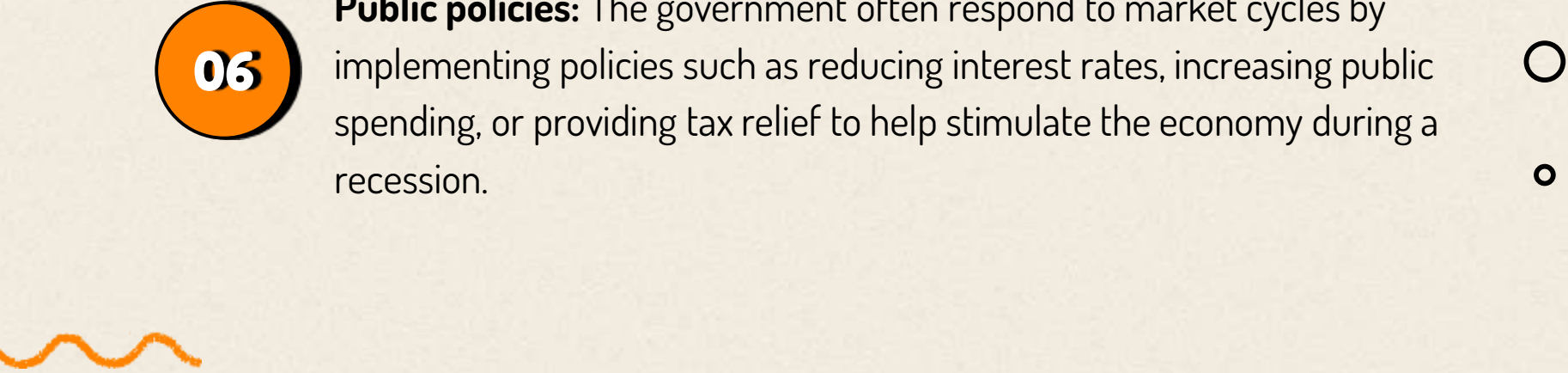


Impact of Market Cycle



05

Housing market: The housing market is often closely tied to the overall economic conditions, so during a market upturn, housing prices may rise, while during a downturn, they may fall.



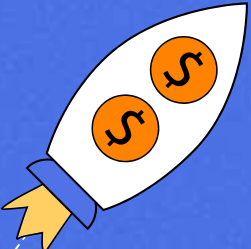
06

Public policies: The government often respond to market cycles by implementing policies such as reducing interest rates, increasing public spending, or providing tax relief to help stimulate the economy during a recession.

Impact of Market Cycle

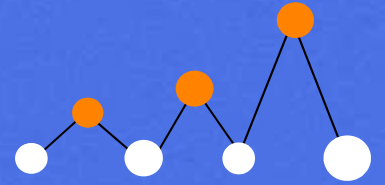
07

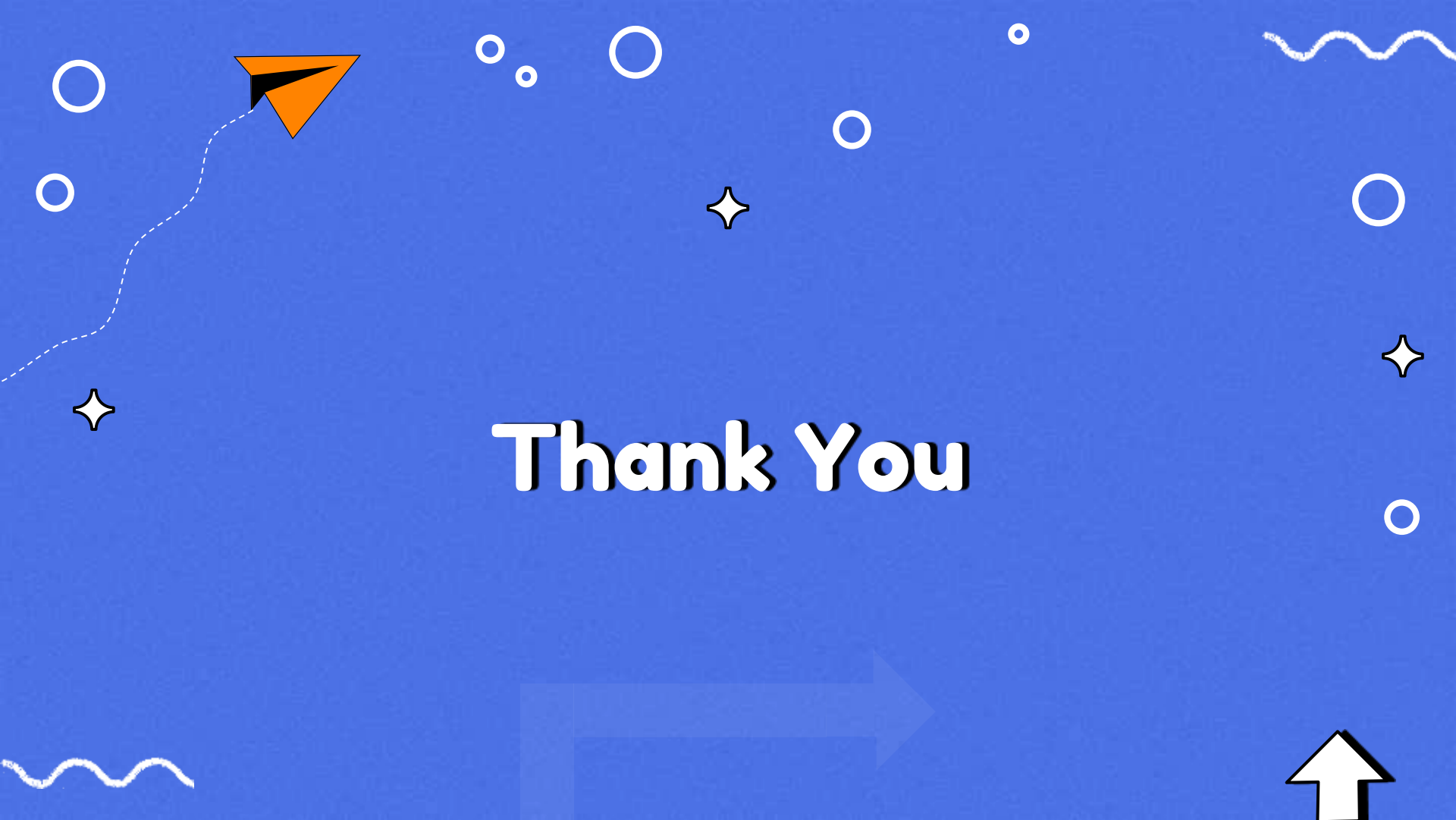
Social and psychological impact: Economic downturns can have a negative impact on the psychological well-being of individuals and communities, as people may feel the stress of losing their jobs, their home, or their savings



"The market can remain irrational longer than you can remain solvent."

- John Maynard Keynes



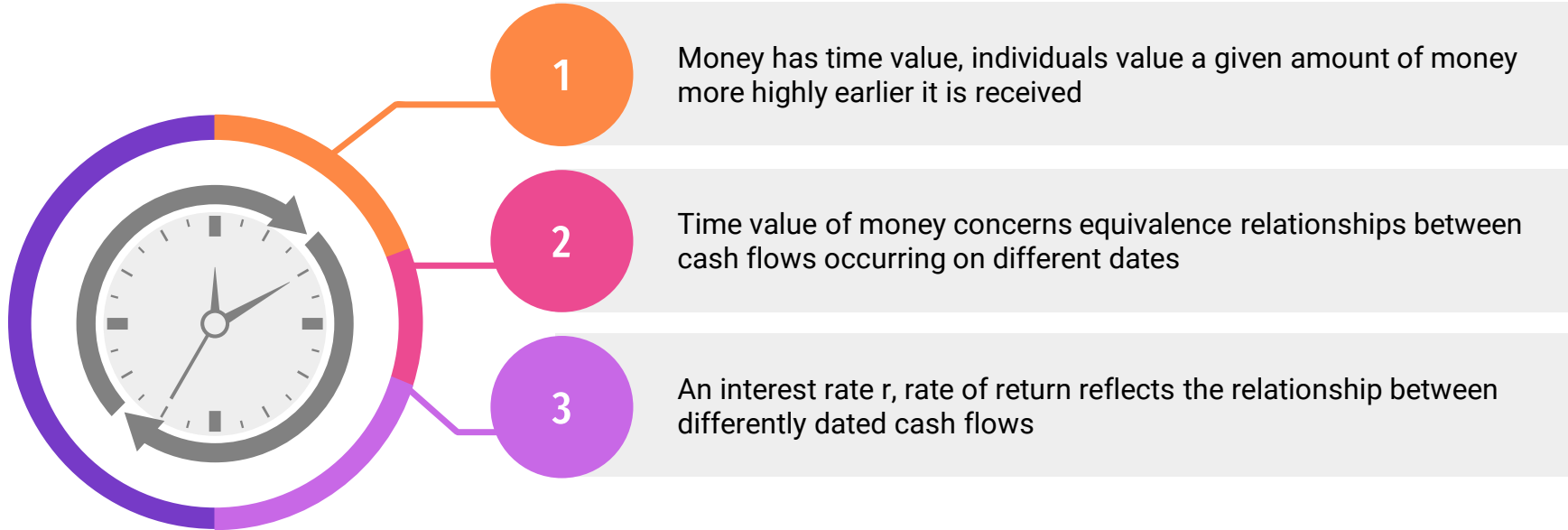


Thank You

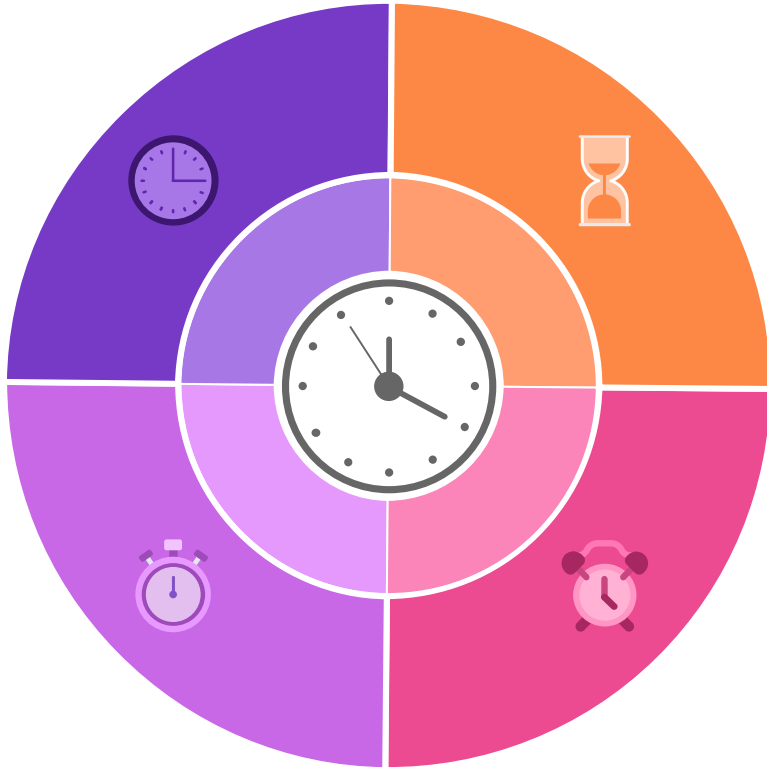
Time Value of Money



Time Value of Money



Simple Interest



Year 1: 5% of $100 = 5 + 100 = 105$

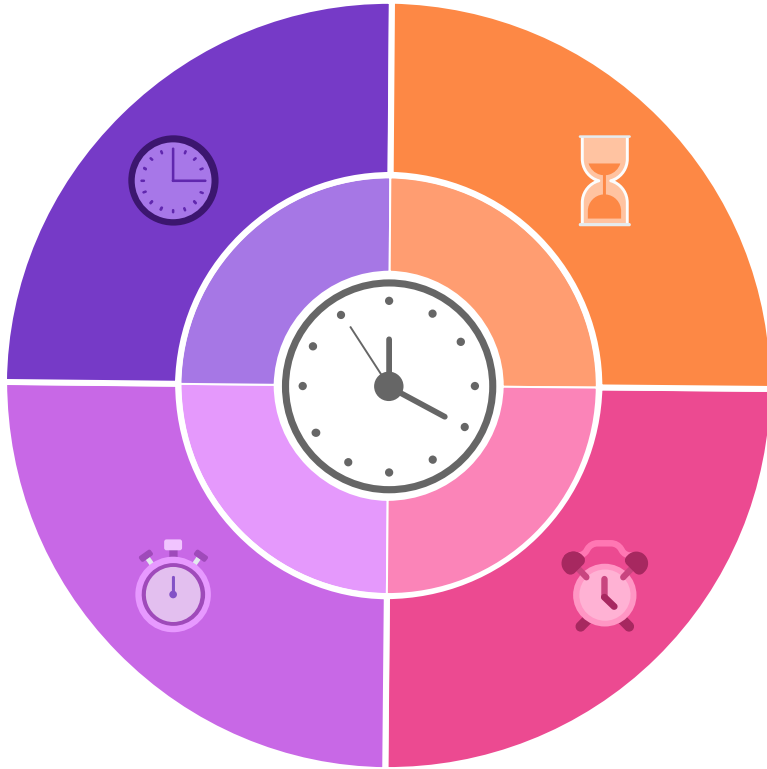
Year 2: 5% of $100 = 5 + 105 = 110$

Year 3: 5% of $100 = 5 + 110 = 115$

Year 4: 5% of $100 = 5 + 115 = 120$

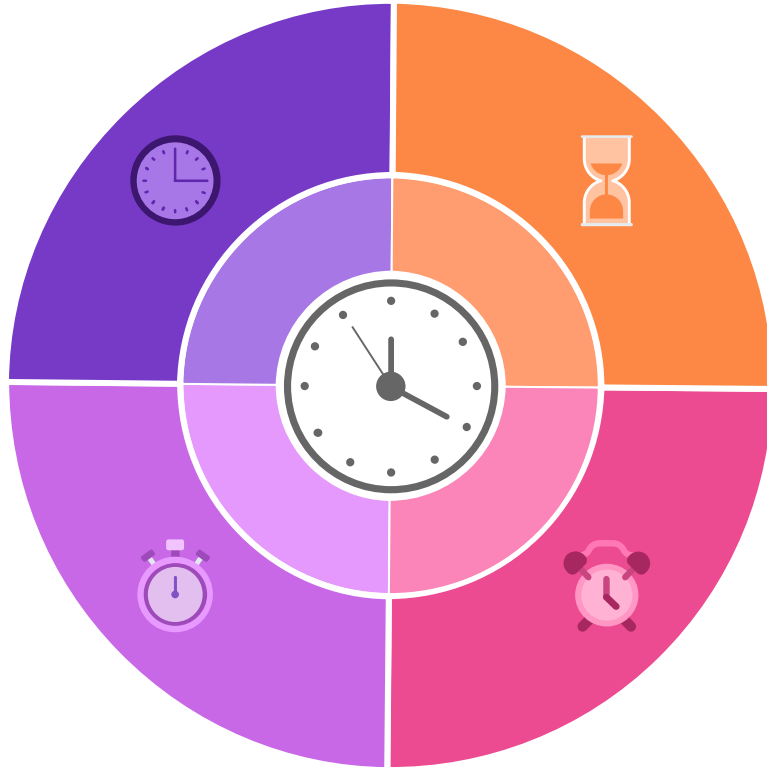
Year 5: 5% of $100 = 5 + 120 = 125$

Compound Interest



**Year 1: 5% of $100.00 = 5.00 + 100.00 =$
 105.00 Year 2: 5% of $105.00 = 5.25 + 105.00$
 $= 110.25$ Year 3: 5% of $110.25 = 5.51 + 110.25 =$
 115.76 Year 4: 5% of $115.76 = 5.79 + 115.76 =$
 121.55 Year 5: 5% of $121.55 = 6.08 + 121.55 =$
 127.63**

Time Value of Money



Single Cash Flow:

$$FV_n = PV(1+r)^n$$

PV=present value of the investment

FV_n=future value of the investment n periods from today

r=rate of interest per period

$$PV = FV_n / (1+r)^n$$

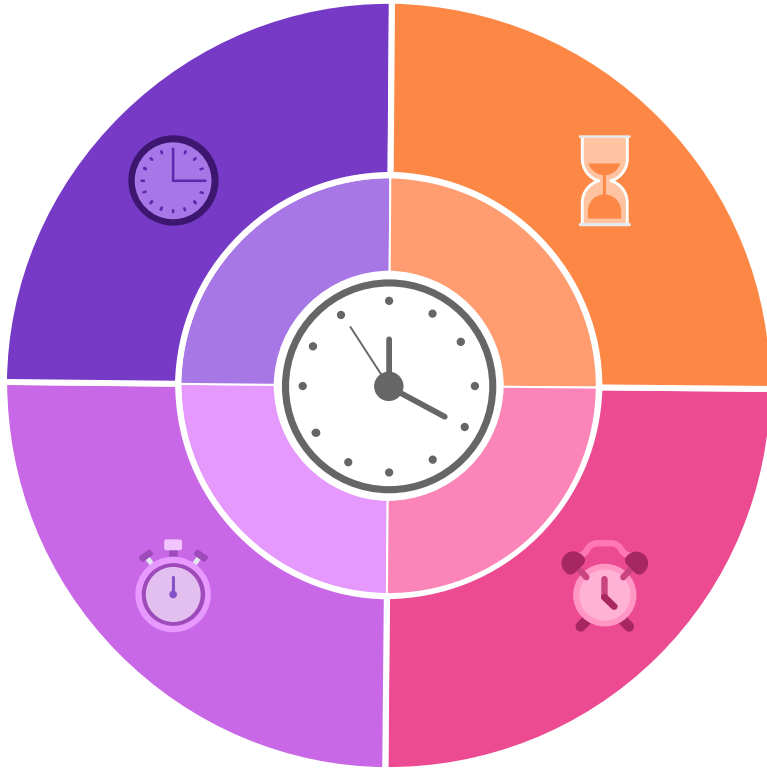
Frequency of Compounding



Let annual interest rate is r
number of compounding periods per year be m

$$\text{Effective annual rate} = (1 + r/m)^m - 1$$

Perpetuity



Present value of an infinite series of equal cash flows

$$PV = A \sum_{t=1}^{\infty} 1/(1+r)^t$$

- Sum of present value converges to:

$$PV = A/r$$

Q1

You invest Rs.25,000 in a 5 year FD at 8% annual interest rate. What is the future value of your investment after 1 year?





Q1

You invest Rs.25,000 in a 5 year FD at 8% annual interest rate. What is the future value of your investment?

ANS

$$PV = \text{Rs.}25,000$$

$$r = 8\% = .08$$

$$n = 5$$

$$FV_n = PV(1+r)^n = 25,000 (1.08)^5 = \text{Rs.}36,733.2$$



Q2

Suppose you invest Rs.10,000 per year in a stock index fund for the next 10 years. Assuming you will earn 12% a year, how much money will you have available after 10 years?



Q2

Suppose you invest Rs.10,000 per year in a stock index fund for the next 10 years. Assuming you will earn 12% a year, how much money will you have available after 10 years?

ANS

$$A = \text{Rs.}10,000$$

$$r = 12\% = .12$$

$$n = 10$$

$$FV_n = \sum_{t=1}^n A(1+r)^{t-1} = \text{Rs.}175,487.35$$



Q3 You want to buy a house 5 years from now. It's expected cost is Rs.50,00,000 five years later. Given interest rate is 8% compounded quarterly, how much should you invest today?



Q3

You want to buy a house 5 years from now. It's expected cost is Rs.50,00,000 five years later. Given interest rate is 8% compounded quarterly, how much should you invest today?

ANS

$$FV = \text{Rs.}50,00,000$$

$$r = 8\% = .08 \text{ compounded quarterly}$$

$$n = 5$$

$$PV = FVn / (1+r/m)^{mn} = 50,00,000 / (1+(.08/4))^{4*5} \\ = \text{Rs.}33,64,856.67$$



Q4

Consider a perpetuity of Rs.100 per year with first payment being made at the beginning of 5th year. What is its present value, given 10% discount rate?

ANS

$$A = \text{Rs.100}$$

$$r = 10\% = .10$$

$$\text{Value after 4 years} = A/r = 100/.1 = 1000$$

$$n = 4$$

$$PV = FV_n / (1+r)^n = 1000 / (1+.1)^4 = \text{Rs.683.01}$$



Grid Trading

ALGO

Grid Trading Algorithm

Grid trading is an algorithmic trading strategy that involves placing buy and sell orders at regular intervals above and below a set price level. The goal of the strategy is to profit from price fluctuations within a range-bound market

In a grid trading strategy, the trader places multiple buy and sell orders at predetermined price levels, called "grid levels," above and below the current market price.

Grid Trading Algorithm

When the market price moves in one direction, the grid trading algorithm automatically executes buy or sell orders at each grid level, depending on the direction of the price movement. For example, if the price rises, the algorithm executes the sell orders, and if the price falls, it executes the buy orders

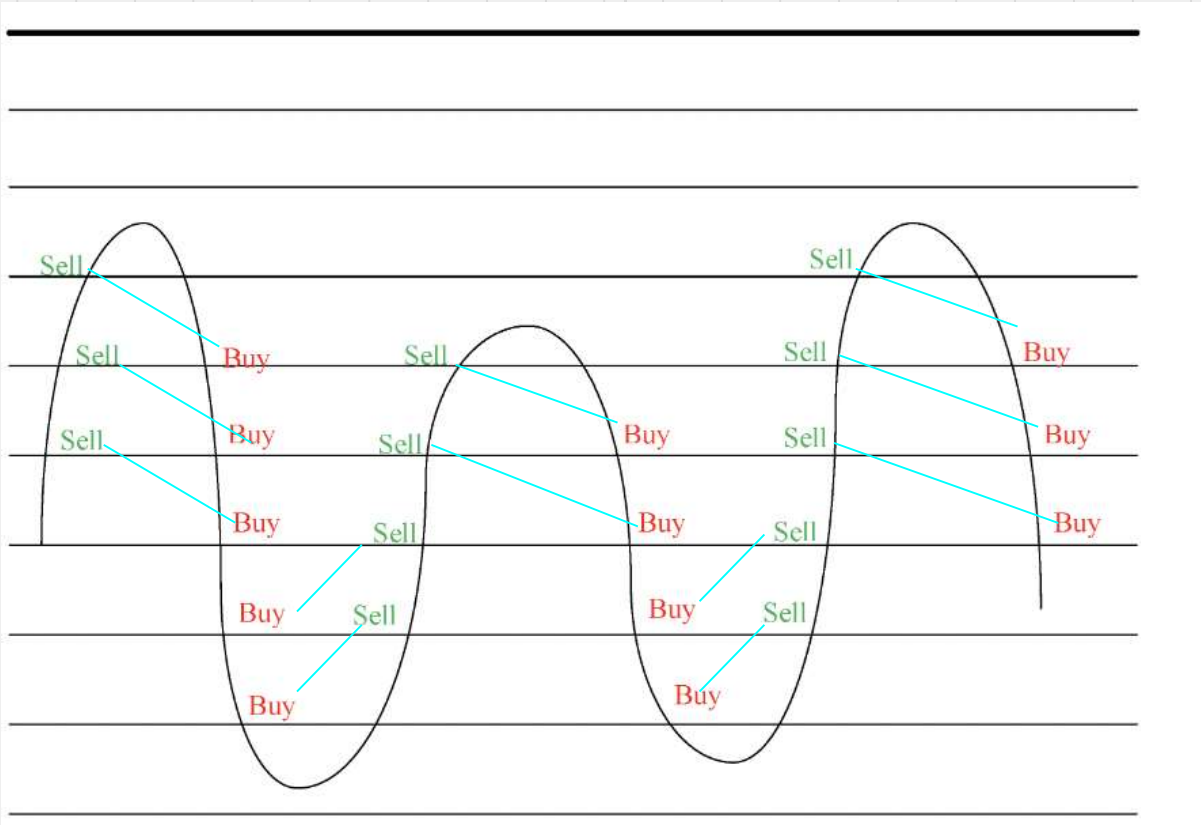
As the price moves up and down, the trader makes profits on the trades executed at each grid level, as long as the price remains within the grid range. The profits from winning trades compensate for the losses from losing trades, resulting in a net profit over time.

Grid Trading Algorithm

One of the key benefits of grid trading is that it can be used in both trending and range-bound markets, making it a flexible trading strategy.

However, it is important to note that grid trading involves a high degree of risk, as the trader is essentially placing multiple trades at once and is exposed to a potential loss if the market moves outside of the grid range

Grid Trading Algorithm





Code in JS



01.

Price Range, Grids

5-20\$, 100 grids

*

02.

Price

We can fetch Price from livefeed, for start we assume price is 12\$

03.

Print

Grid array, buying trades, selling trades

*

04.

Update

Grid, calculate wallet Sum on Price change
12->10->13

